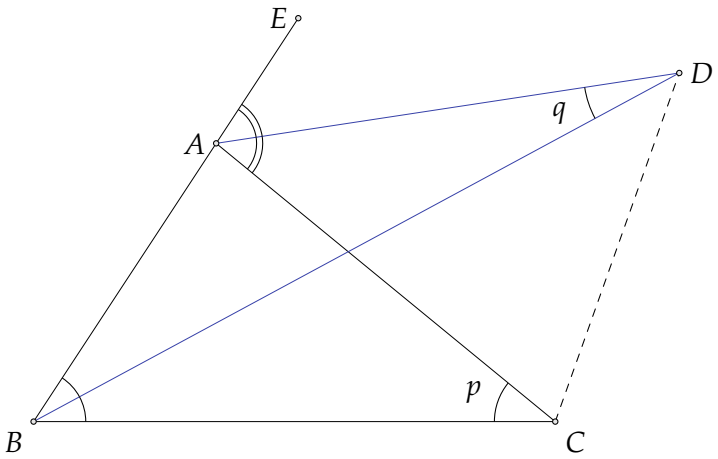
Cayley Olympiad 2023, Problem 2: four scripts, four marks

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The problem

In triangle ABC, the side BA is extended beyond A to a point E. The bisector of angle ABC and the bisector of angle EAC meet at D. Let $\angle ACB = p$ and $\angle ADB = q$. Prove that $p = 2q$.



Write $\angle ABD = \angle DBC = a$ and $\angle EAD = \angle DAC = b$.

Script A **10/10**

Since BD bisects angle ABC and AD bisects angle EAC, $\angle ABD = \angle CBD$ and $\angle EAD = \angle CAD$. Let the two angles be a and b respectively; then $\angle ABC = 2a$ and $\angle EAC = 2b$. Since an exterior angle of a triangle is equal to the sum of the other two interior angles, we may deduce that $\angle EAC = \angle ACB + \angle ABC = p + 2a$, and similarly $\angle EAD = \angle ADB + \angle ABD = q + a$. So $2b = p + 2a$ and $b = q + a$. Then $2 \times (2) - (1)$ gives $2q - p = 0$, so $p = 2q$.

Marker's comment. Complete. Each exterior angle equation comes with the angles it concerns and the elimination is shown. The contest script wrote $\angle ACD$ for $\angle ACB$ in the first exterior angle equation (corrected above), a slip that costs nothing because the equation makes the meaning clear.

Script B **7/10**

Let $\angle ABD = \angle DBC = a$ and $\angle EAD = \angle DAC = b$. Then $2b = p + 2a$ and $b = q + a$, so $p = 2q$.

Marker's comment. The right argument, but neither equation has a reason. The first is the exterior angle of triangle ABC at A, and the second is the exterior angle of triangle ABD at A, which needs E, A, B collinear. Saying so and writing out the elimination gives full marks.

Script C 4/10

Let $\angle ABD = \angle DBC = a$ and $\angle EAD = \angle DAC = b$. In triangle ABC, $2b = p + 2a$ by the exterior angle. In triangle ADC, the exterior angle at D gives $b = q + a$. Subtracting twice the second equation from the first gives $p = 2q$.

Marker's comment. The set-up and the first equation earn the marks. The second equation is true but the reason is false, since $\angle ADB$ is not an exterior angle of triangle ADC. A true statement with a wrong reason scores nothing. The correct triangle is ABD, with exterior angle $\angle EAD$.

Script D 1/10

Take $\angle ABC = 60^\circ$ and $\angle ACB = 40^\circ$. Then $\angle BAC = 80^\circ$, so $\angle EAC = 100^\circ$ and $\angle DAC = 50^\circ$. In triangle ABD, $\angle BAD = 80^\circ + 50^\circ = 130^\circ$ and $\angle ABD = 30^\circ$, so $\angle ADB = 20^\circ$. Hence $\angle ACB = 2\angle ADB$.

Marker's comment. The arithmetic is correct, but one triangle proves nothing about the others. The mark is for understanding the configuration. The same working with $2a$ and p in place of 60° and 40° is a proof.

What full marks needs

Each exterior angle equation must name its triangle, and the elimination must be shown. Script C is as long as Script A and loses six marks by taking the second equation from triangle ADC.